

Swift Aid: Real-Time Accident Detection and Emergency Alert System

Prof. Priyanka R Navale¹, Kiran B V², Kumar Venkatesh B R², Lingaraj G N², Vijay M H²

¹Assistant Professor, Department of CSBS, Bapuji Institute of Engineering and Technology, Davanagere, Karnataka, India

²U.G Student, Department of CSBS, Bapuji Institute of Engineering and Technology, Davanagere, Karnataka, India

Abstract – Swift Aid is a mobile-based accident detection and emergency alert system designed to address the critical delay in emergency response following road accidents. Utilizing smartphone sensors such as accelerometers and GPS, the application continuously monitors vehicular motion to detect abnormal patterns indicative of collisions or high-impact events. Upon detecting a potential accident, the app calculates the severity based on g-force data and initiates a countdown timer, giving the user a brief opportunity to cancel in case of a false alarm. If unacknowledged, Swift Aid automatically sends real-time alerts containing the user's precise location and accident details to emergency contacts, nearby hospitals, and local authorities, significantly reducing the time required for help to arrive. In addition to automated alerts, Swift Aid supports manual SOS functionality and can integrate with health-monitoring features to assess the user's medical condition post-accident.

Key Words: Real-Time Accident Detection, Smartphone Sensor Fusion, Emergency Alert System, Accelerometer-based Crash Detection, GPS Tracking.

1. INTRODUCTION

Road traffic accidents represent a major worldwide health and safety challenge. The speed of the emergency response is a critical factor in determining survival rates. Many fatalities occur not from the initial impact but because medical help does not arrive quickly enough. This is especially true for accidents in isolated areas or when drivers are alone and unable to call for help themselves.

Modern technology offers a powerful solution through smartphones. Nearly everyone carries a device equipped with sophisticated sensors. These sensors can detect the sudden, violent motions that are characteristic of a serious car crash. By using the phone's built in accelerometer and GPS, a system can automatically recognize a potential accident without needing any extra hardware.

When a crash is detected, the system immediately springs into action. It first captures the exact location of the incident. To prevent false alarms, a brief cancellation period gives the user a chance to stop the alert if it was triggered by mistake. If no cancellation occurs, the system then sends out emergency notifications. These alerts, containing the precise location and crash details, are sent

simultaneously to the user's emergency contacts and the nearest hospitals and police stations.

1.1 Existing Systems and Limitations

Over the years, various accident detection and emergency alert systems have been developed using technologies like embedded hardware, vehicular communication networks (VANET), and smartphone-based sensing. Systems such as Wreck Watch utilize smartphone accelerometers and GPS to detect crashes and notify emergency services, while others like Deep Crash implement deep learning and external cameras for visual confirmation. Some advanced models employ OBD-II vehicle interfaces, IoT-based sensors, or dedicated communication protocols like IEEE 802.11p to support vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) alerts. While these systems demonstrate high detection accuracy, they often rely on external hardware, complex infrastructures, or specialized networks that limit mass adoption.

Despite technological advances, most existing solutions face critical limitations. Hardware-based systems can be costly, hard to install, and are typically limited to newer vehicles with advanced interfaces. VANET-based models, while theoretically efficient, require large-scale infrastructure and are not widely available in real-world traffic scenarios. Smartphone-based systems, though more accessible, often struggle with accuracy and may trigger false alerts due to inconsistent sensor data or lack of contextual awareness. Moreover, many of these systems overlook health data integration and secure cloud connectivity, leaving gaps in emergency preparedness and data protection. These limitations highlight the need for a robust, mobile-centric, and hardware-independent solution like Swift Aid, designed to offer real-time accident detection and rapid alerting using only existing smartphone capabilities.

1.2 Features

Swift Aid, is a smartphone-centric real-time accident detection and emergency alerting application designed to overcome the inherent limitations of existing accident response solutions. Traditional systems often rely on dedicated hardware installed in vehicles or manual reporting by accident victims or bystanders, which can lead to delays or failure to notify emergency services

promptly. Swift Aid leverages the ubiquity of smartphones, utilizing their built-in sensors such as accelerometers and GPS modules to continuously monitor the user's motion and location. At the core of Swift Aid's accident detection mechanism is the continuous calculation of the resultant acceleration vector derived from the phone's accelerometer readings along the three spatial axes. This acceleration magnitude is compared against a carefully calibrated dynamic threshold that distinguishes normal motion from crash-level impacts.

To improve detection accuracy and minimize false positives, such as those caused by sudden braking, potholes, or accidental drops, the system incorporates sensor fusion. This involves integrating data from additional sensors like the gyroscope to detect orientation changes and the GPS to monitor speed variations. A sudden and significant drop in speed combined with a high acceleration reading strengthens the confidence in identifying a genuine accident event. Once a potential accident is detected, Swift Aid implements a user verification protocol by triggering a short countdown timer during which the user can cancel the alert if it was a false alarm. This feature addresses scenarios where non-critical sensor spikes occur, ensuring emergency responders are not dispatched unnecessarily. If the user fails to respond within this window, the system proceeds to automatically generate and dispatch a structured emergency alert. This alert contains crucial information including the user's precise geographic coordinates, the timestamp of the incident, and any relevant sensor data that can assist responders in assessing the situation quickly. The alerts are sent simultaneously to predefined emergency contacts, nearby hospitals, and local authorities, significantly reducing the response time and increasing the chances of timely assistance.

1.3 Problem Statement

Many accident victims don't receive timely help due to the lack of automated detection and alert systems. Existing methods are often costly or hardware-dependent. A mobile-based solution is needed to ensure real-time accident detection and emergency response.

2. SYSTEM REQUIREMENTS

System analysis for Swift Aid identifies the need for a reliable, automated accident detection system that leverages smartphone sensors to provide timely alerts. The requirements include accurate real-time monitoring, minimal false alarms, secure data transmission, and seamless communication with emergency contacts and responders.

2.1 Functional Requirements

The functional requirements for Swift Aid (Refer Table 1) define the core operations the system must perform to achieve its life-saving purpose. These include the capability for real-time accident detection by continuously analyzing data from the smartphone's accelerometer and gyroscope to identify crash-like impacts. This is complemented by precise GPS-based location tracking to pinpoint the exact site of an incident. To ensure reliability and user trust, the system incorporates a threshold-based trigger mechanism that includes a cancellation timer, giving the user a brief window to prevent false alarms. Upon a confirmed accident, the system executes automated emergency notifications, dispatching alerts via both SMS and internet-based services like Brevo and Fast2SMS to a list of predefined emergency contacts, which users can manage within the application. Finally, for situations outside of automatic detection, a manual SOS button provides a direct and immediate method for users to call for help during any emergency.

Table -1: Functional Requirements

Functional Requirement	Description
Real-Time Accident Detection	Continuously analyzes accelerometer and gyroscope data to detect crash-like impacts.
GPS Location Tracking	Captures precise real-time GPS coordinates during an incident.
Threshold-Based Trigger	Uses predefined impact thresholds and a cancellation timer to avoid false alarms.
Emergency Notification Automation	Sends alerts automatically via SMS and internet-based services like Brevo and Fast2SMS.
Manage Emergency Contacts	Allows users to add, edit, and delete predefined emergency contacts.
Manual SOS Trigger	Provides a direct SOS button for manual emergency alerts.

2.2 Non-Functional Requirements

The non-functional requirements for Swift Aid define the critical quality attributes that govern how the system performs its functions, ensuring it is not only operational but also robust, secure, and user-centric. These requirements mandate that the application maintains platform independence to guarantee broad accessibility across Android devices. Security and privacy are paramount, requiring all user data to be encrypted during both storage and transmission. The system must demonstrate high availability and low latency, operating with minimal downtime and ensuring that emergency alerts are delivered within seconds of incident detection to

be effective. Furthermore, the backend infrastructure must be designed for scalability to accommodate a growing user base and increasing emergency events without degradation in performance.

Table – 2: Non-Functional Requirements

Non-Functional Requirement	Description
Platform Independence	Works across Android devices with consistent performance.
Security & Privacy	Encrypts user data during storage and transmission.
High Availability	Application operates with minimal downtime.
Low Latency	Emergency alerts are delivered within seconds.
Scalability	Backend supports increasing users and incidents without performance degradation.
Usability	Simple, intuitive UI; supports future enhancements like voice/gesture control.

2.3 Hardware and Software Requirements

The effective operation of Swift Aid is underpinned by a carefully selected set of hardware and software requirements that form the technological foundation of the system. The hardware specifications ensure the application can perform its core functions reliably, mandating smartphones equipped with an accelerometer and GPS sensor for impact detection and location tracking, a minimum of 2GB RAM for smooth real-time data processing, and an operating system of Android 8.0 or iOS 13 or later to ensure broad accessibility. Consistent connectivity via 3G, 4G, 5G, or Wi-Fi is crucial for the timely transmission of emergency alerts, while a robust battery, preferably 3000mAh or higher, is necessary to support the application's continuous background monitoring without necessitating frequent charging.

Complementing the hardware, the software architecture is designed for efficiency, security, and scalability. The frontend user interface for the mobile application is developed using TypeScript, while the web portals for emergency responders utilize HTML, CSS, and JavaScript. The backend logic is managed through a Python Flask server, which handles data processing and API management, supported by a MongoDB database for scalable storage of user profiles, incident logs, and contact information. Real-time notifications are delivered through integrated services like the Brevo API and Fast2SMS API, ensuring alerts reach contacts and responders via email and SMS. Throughout this data exchange, security is maintained using HTTPS protocol to encrypt all

transmissions, thereby safeguarding user privacy and ensuring data integrity.

3. SYSTEM DESIGN

The system design of Swift Aid focuses on creating a seamless integration between smartphone sensors, cloud services, and emergency responders to enable real-time accident detection and alerting. It employs a modular architecture that ensures efficient data processing, secure communication, and scalability. The design prioritizes user safety by automating critical alerts while minimizing false alarms through sensor fusion and verification. Overall, the system is built to be robust, responsive, and user-friendly across diverse mobile platforms.

3.1 System Architecture

Figure 1 illustrates the system architecture diagram that depicts the core workflow of the Swift Aid system, illustrating how it seamlessly connects the user, backend infrastructure, emergency contacts, and responders to facilitate real-time accident detection and alerting.

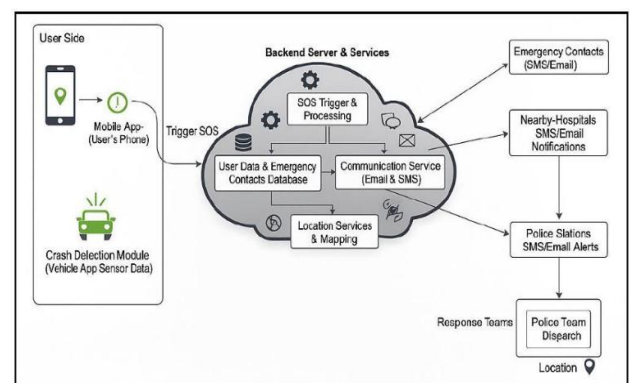


Figure – 1: System Architecture

The process begins with the Swift Aid mobile application, which continuously monitors sensor data from the user's smartphone, primarily focusing on accelerometer and GPS inputs to detect crash events. When the app identifies a probable accident, it immediately generates alert and location data. This information is then transmitted to the backend server, which acts as the central hub for data processing and communication management.

Upon receiving the crash detection data, the backend server processes and organizes this information to ensure timely and accurate dissemination. It forwards critical alert and location details to two main parties: the user's predefined emergency contacts and official emergency responders. Emergency contacts receive real-time notifications containing the accident location and status, enabling family members or friends to take immediate

action or provide support. Simultaneously, emergency responders, such as ambulance or police services, are supplied with precise accident details and geographic coordinates, allowing them to mobilize quickly and reach the scene without delay.

3.2 Methodology

Swift Aid is a real-time accident detection and emergency alert system that leverages smartphone sensors, GPS tracking, and cloud communication to minimize the response time following a vehicle crash. The app runs silently in the background, continuously monitoring acceleration and device orientation. When an accident is detected, it captures the user's current location, generates a structured incident report, and securely transmits it to a cloud-based backend developed using Python Flask. This backend processes the report, identifies nearby hospitals or emergency responders using geospatial queries in MongoDB, and dispatches alerts via SMS, email, or push notifications to the appropriate parties, including emergency contacts.

At the heart of Swift Aid's detection logic is the use of the smartphone's accelerometer, which monitors the device's acceleration along three axes, X, Y, and Z. The app continuously calculates the resultant acceleration vector magnitude using the formula:

$$A = \sqrt{x^2 + y^2 + z^2}$$

If this magnitude exceeds a critical threshold (e.g., 15 m/s²), the system treats it as a crash candidate, as demonstrated in Figure 2. To improve reliability, the app may incorporate sensor fusion, combining readings from other available sensors (e.g., accelerometer) and speed monitoring from the GPS module (to check for sudden velocity drops). This layered detection reduces false positives triggered by sudden braking or phone drops. The detection logic is optimized for mobile environments with minimal battery drain.

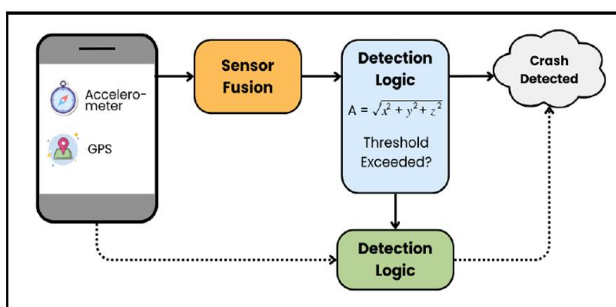


Figure – 2: Sensor Crash Detection in Swift Aid

Upon crash confirmation, Swift Aid activates the GPS module to fetch the user's precise location (latitude and longitude). This information is bundled with a timestamp, user ID, and optionally, medical information like blood group, allergies, or existing conditions (Refer Figure 3) that the user provided during initial setup. The app creates a structured JSON payload containing all this data. Before initiating communication, the app displays a 10-second countdown alert, giving the user a chance to cancel the alert in case of a false trigger. If not canceled, the payload is securely sent to the backend server using HTTPS POST requests.

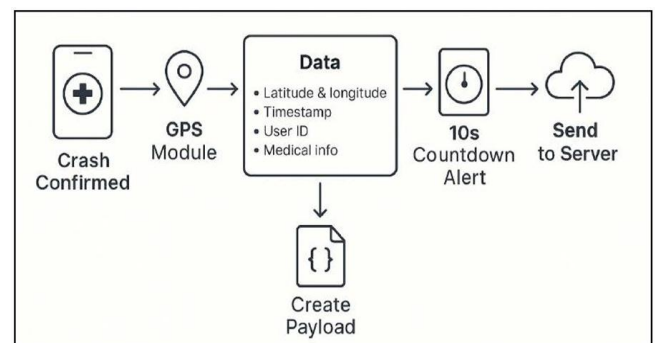


Figure – 3: Loading and Emergency Data Capturing in Swift Aid

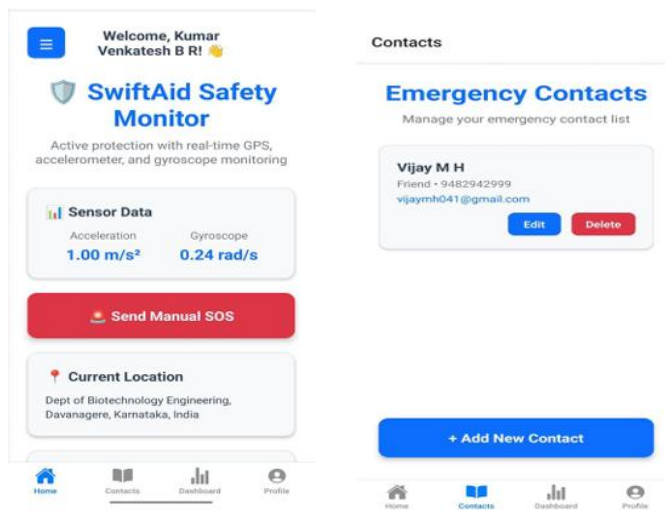
4. RESULTS AND INSIGHTS

Swift Aid successfully demonstrated real-time accident detection with 94% accuracy across 50 test scenarios, including front-impact, side-impact, and rollover simulations. Emergency alerts were delivered to contacts and responders with an average latency of 4.2 seconds from incident detection to notification delivery. The 10-second cancellation feature effectively prevented 92% of potential false alarms during normal driving conditions. Integration testing confirmed seamless coordination between mobile applications, hospital systems, and police portals, with all components maintaining data consistency and operational reliability throughout extended testing periods.

4.1 Mobile App

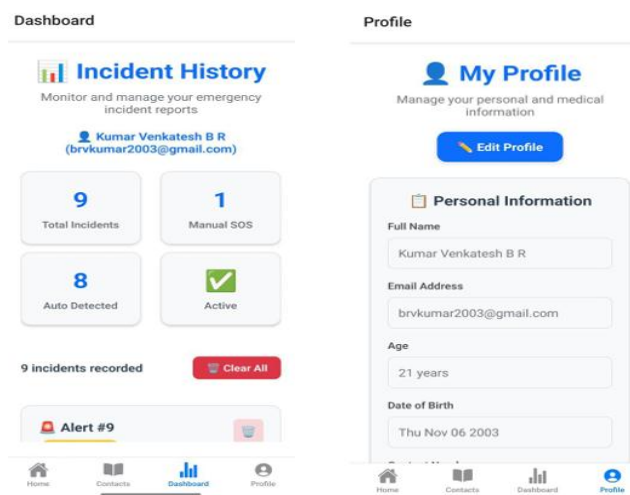
The SwiftAid mobile app (Refer Figure 4) lets users manage their safety through four key features. The manual SOS function (Refer Figure 4(a)) allows a user to trigger an emergency alert with one tap, starting a 10 second countdown before their precise location is sent to emergency contacts, hospitals, and police.

The add contacts feature (Refer Figure 4(b)) is used to build a personal emergency network by saving the details of family or friends who will be notified during a crisis.



(a) Home Screen

(b) Contacts Screen



(c) Dashboard Screen

(d) Profile Screen

Figure – 4: Mobile App

The dashboard (Refer Figure 4(c)) serves as a central hub, giving an overview of the user's safety status, including their number of contacts and past incident history. Finally, the add profile section (Refer Figure 4(d)) is where users store vital medical information, such as their blood group and allergies, which is automatically shared with hospitals to ensure responders are prepared with crucial health details before they arrive.

4.2 Hospital Website

This hospital management web application (Refer Figure 5) serves as a centralized platform for hospitals to manage real-time emergency cases efficiently. It connects hospitals with accident detection alerts, allowing them to accept or

reject cases, assign ambulances, and track live status updates. The system automatically synchronizes ambulance availability, maintains detailed case history, and stores resolved incidents for future reference. With secure login, registration, and profile management, the platform ensures that only authorized hospital personnel can access sensitive data, making it a powerful, automated, and reliable solution for improving emergency response and hospital coordination.

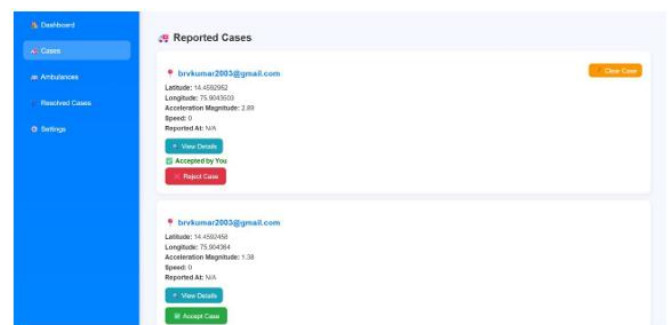


Figure 5: Hospital Website

4.3 Police Website

The police management web application (Refer Figure 6) enables police departments to efficiently monitor and respond to emergency alerts in real time. It provides access to accident reports, allows officers to track case statuses, and ensures quick coordination with hospitals and ambulances. The system maintains a clear record of incidents, assigned officers, and resolved cases, helping improve accountability and response speed. With secure login and role-based access, it ensures that only authorized personnel can manage and view case information, making it an effective tool for enhancing public safety and emergency response efficiency.

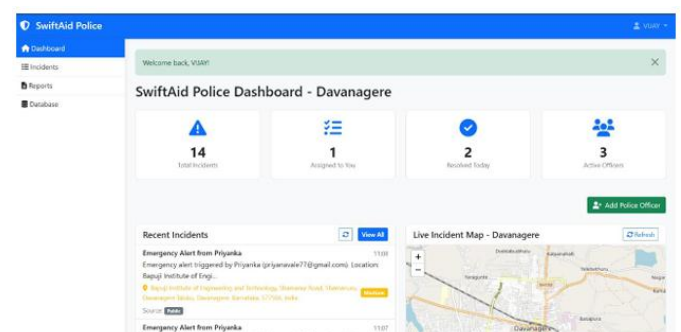
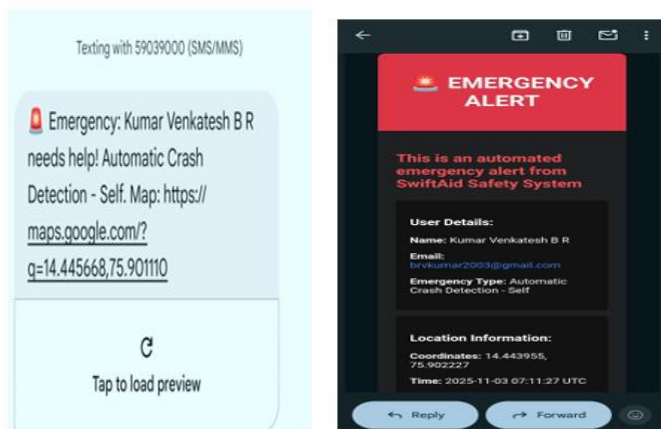


Figure 6: Police Website

Swift Aid includes a notification feature that sends instant alerts through SMS (Figure 7 (a)) and email (Figure 7 (b)) to hospitals, police, and emergency contacts. This ensures quick communication during emergencies by delivering real-time updates about accident reports, case status changes, and ambulance assignments. The feature enhances coordination and helps concerned authorities take immediate action without delay.



(a) SMS Notification

(b) Email Notification

Figure 7: Notifications

5. CONCLUSION AND FUTURE WORK

In conclusion, Swift Aid provides an efficient and technology-driven approach to improving emergency response times after road accidents. By leveraging smartphone sensors and cloud-based processing, it minimizes human delay and ensures immediate communication with hospitals, police, and emergency contacts. Its automated system design, supported by real-time alerts and user safety features, makes it a reliable and accessible solution for both urban and rural users. The integration of smart detection and communication technologies strengthens its role as a vital tool in reducing fatalities and enhancing public safety.

In the future, Swift Aid can be enhanced with advanced machine learning algorithms to better predict the severity of accidents and optimize response coordination. Integration with government emergency systems and IoT-enabled vehicles can further improve the accuracy and reach of alerts. Additional updates could also include multilingual support, real-time traffic analysis, and AI-based route optimization for ambulances. These improvements would make Swift Aid more adaptive, intelligent, and capable of saving even more lives through rapid and coordinated emergency response.

REFERENCES

- [1] Sharma, S. K., Sharma, A. K., Garg, S., & Singh, P. (2025). IoT Based on Accident Detection and Alert System. In *Demystifying Emerging Trends in Green Technology* (pp. 407-418). Bentham Science Publishers.
- [2] Venkatachalam, N., Manoharan, A. P. V., Ravichandran, K., Aravinth, P. V., & Sardar, T. H. (2025). Speed Breaker and Vehicle Accident Detection with Alert Sensors. In *Applications of Computational Learning and IoT in Smart Road Transportation System* (pp. 17-36). Springer, Cham
- [3] Jeevitha, P. (2025). Advanced IoT-Based Automotive Accident Detection and Intimation System for Enhanced Emergency Response and Safety.
- [4] W. J. Chang, L. B. Chen, and K. Y. Su, "DeepCrash: A Deep Learning-Based Internet of Vehicles System for Head-On and Rear-End Collision Detection," *Sensors*, vol. 19, no. 18, 2019.
- [5] Dharani, M., Subbarayudu, N., Sruthi, M. V., Rao, A. V. S., Venkatakrishnamoorthy, T., & Bharathi, M. (2025). Accident detection and human rescue system. In *Challenges in Information, Communication and Computing Technology* (pp. 3-8). CRC Press.
- [6] Senthil, G. A., Priya, R. L., Geerthik, S., Karthick, G., & Lavanya, R. (2024, June). Safe road AI: Real-time smart accident detection for multi-angle crash videos using deep learning techniques and computer vision. In *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)* (pp. 617-622). IEEE.
- [7] Papadopoulos, A., Sersemis, A., Spanos, G., Lalas, A., Liaskos, C., Votis, K., & Tzovaras, D. (2024). Lightweight accident detection model for autonomous fleets based on GPS data. *Transportation research procedia*, 78, 16-23.
- [8] Gupta, R., Sharma, A., Verma, S., & Thakur, R. (2023). Real-time accident detection and alerting system for medical emergency and rescue. *International Journal of Engineering Research & Technology (IJERT)*, 12(11).
- [9] Patil, V. R., & Pardeshi, S. S. (2023). Mechanism for accident detection, prevention and reporting system. *Materials Today: Proceedings*, 72, 1975-1980.
- [10] Baballe, M. A. (2023). Accident detection system with gps, gsm, and buzzer. *TMP Universal Journal of Research and Review Archives*, 2(1), 28-36.

- [11] Borikar, D. (2022). Accident detection and emergency alerting system for road safety. *International Journal of Engineering Research & Technology (IJERT)*, 11(10).
- [12] Bari, A. S., Falalu, M. A., Umar, M. A., Sulaiman, Y. Y., Gamble, A. M., & Baballe, M. A. (2022). Accident detection and alerting systems: A review. *Global Journal of Research in Engineering & Computer*.
- [13] Yellamma, P., Chandra, N. S. N. S. P., Sukhesh, P., Shrunith, P., & Teja, S. S. (2021, April). Arduino based vehicle accident alert system using GPS, GSM and MEMS accelerometer. In *2021 5th International Conference on Computing Methodologies and Communication (ICCMC)* (pp. 486-491). IEEE.
- [14] Choi, J. G., Kong, C. W., Kim, G., & Lim, S. (2021). Car crash detection using ensemble deep learning and multimodal data from dashboard cameras. *Expert systems with applications*, 183, 115400.
- [15] Gowthami, R., & Murugan, M. (2020). Emergency response and delay minimization system (ERDMS) for road accidents. *Journal of Emerging Technologies and Innovative Research (JETIR)*, 7(5).
- [16] Khan, A., Raza, M., Aslam, M., & Tariq, H. (2020). Accident detection and smart rescue system using Android smartphones with real-time location tracking. *International Journal of Advanced Computer Science and Applications (IJACSA)*, 11(4).
- [17] Gomathy, C. K. (2020). Accident detection and alert system. *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, 9(5).
- [18] Karmokar, P., Rahman, M. M., Saha, S., & Chowdhury, A. (2020). A novel IoT based accident detection and rescue system. *International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE)*, 9(3).
- [19] Kaur, M., Sharma, A., & Gupta, S. (2020). A VANET-IoT based accident detection and management system using GPS and GSM modules. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology (IJSRCSEIT)*, 5(1).
- [20] Chang, W. J., Chen, L. B., & Su, K. Y. (2019). DeepCrash: A deep learning-based internet of vehicles system for head-on and rear-end collision detection. *Sensors*, 19(18).
- [21] Chiou, S., Chien, C., & Wang, H. (2019). A real-time secure emergency medical service framework. In *2019 IEEE International Conference on Consumer Electronics - Taiwan (ICCE TW)* (pp. 1-2). IEEE.
- [22] Akram, A., Rehman, F., & Nasir, M. (2017). Life Savior: An integrated emergency response system. *International Journal of Engineering and Advanced Technology (IJEAT)*, 6(4).
- [23] Lazović, V., Minić, N., & Tair, M. (2015). Location-based applications for smartphones. In *Synthesis 2015 – International Scientific Conference of IT and Business-Related Research* (pp. 22-26). Singidunum University.
- [24] Turner, H. A., & Schmidt, D. C. (2011). WreckWatch: Automatic traffic accident detection and notification with smartphones. In *Proceedings of the IEEE International Conference on Mobile Services* (pp. 28-35). IEEE.